VR-Sets. Preprint.

Part I. Foundations

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I.1. Place in the cycle of works

VR-Sets is an operational theory of sets, the third work in a cycle developing the VR programme:

> **VR. A Formal System** (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20212092) — a formal axiomatisation of arithmetic on three primitives $\{\varnothing, \to, t\}$ and four axioms. It is equivalent to Peano arithmetic and consistent relative to ZF. The basic ontological inventory: the sole primitive $\varnothing$, implication $\to$, and the unary successor operator t.
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> **VR-Numbers** (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20272743) — operational extensions over the natural numbers of VR. The integers $\mathbb{Z}$, rationals $\mathbb{Q}$, reals $\mathbb{R}$, and complex numbers $\mathbb{C}$ are constructed without introducing ordered pairs as objects. In place of "a pair of numbers" — operational procedures with order built into the description itself. Substantively: $\mathbb{R}_{VR}$ is isomorphic to the field of computable reals (a countable subfield of the classical $\mathbb{R}$); the two-axis structure of $\mathbb{C}$ is structurally motivated by the duality of axiom A1 (the two implications $F \to F$ and $F \to \top$).
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> **VR-Sets** — the present work. An operational theory of sets that unifies ZFC and ZFA through reference semantics. It resolves the three open questions left in Part VII.3 of VR-Numbers: an operational reformulation of the ZF axioms, the relation to Tarski's theorem on the undefinability of truth, and the status of the axiom of choice.

The common methodological line of the three works is **ontological minimalism**: a sole primitive $\varnothing$, with everything else as operational actions. The slogan, formulated in VR-Numbers Part VI.5: "only $\varnothing$ is, all else is doing." In VR-Sets this principle is applied to sets: a set *is* its own operationality, not a separate entity standing behind it.

I.2. The central idea

In classical ZF, a set is a container. The relation $x \in y$ is read as "x is located inside y." Paradoxes of self-reference (Russell) and paradoxes of choice (Banach–Tarski) arise from tensions in this container metaphor: a container of "all containers that do not contain themselves" contradicts itself; a container holding "all" elements of a continuum turns out to be a source of non-measurable decompositions.

In VR-Sets, a set is a functionality. A set does not "contain" elements; a set *is* the procedure that, when asked "what are your elements?", either reveals a definite element or reveals nothing. The elements are themselves sets (themselves functionalities). The sole initial entity is $\varnothing$, the empty functionality, which reveals nothing.

The membership relation $x \in y$ is read as reference: x was used in constructing the functionality y. Not as physical containment. This dissolves the paradox of self-reference at the ontological level: $x \in x$ does not mean "x is physically inside itself" but rather "the functionality x, in one of its responses, returns x." This is a definite response, not a structural cycle.

Two ontological modes correspond to two choices regarding self-reference. The ZFC-mode forbids functionalities from referring to themselves in their responses; the ZFA-mode permits this. In both modes a set is operationality; the difference lies in which operationalities are admitted. Part IV of the preprint formalises this distinction.

Summary of the central idea. A set is not a vessel with contents, but a description of behaviour: "here is what I will reveal when asked." Everything else is consequence of this change of metaphor.

I.3. Ontological position

We fix here the ontological commitments inherited in VR-Sets from VR and VR-Numbers.

Position 1: ∅ is the sole primitive

In the ontology of VR-Sets there is only one primary object — ∅, the empty operationality. All other objects are constructions over ∅, realised through operational functionalities. No atoms (in the sense of urelements in ZFA), no individuals, no ordered pairs as independent objects. Anything that is "something else" is an *action*, not a separate entity.

Position 2: all is sets

The elements of sets are themselves sets. This is consistent with Position 1: if elements could be some "atoms" distinct from sets, we would have a second primitive kind of object, violating minimalism. In VR-Sets, the only thing that can be revealed upon querying a membership functionality is another set (another functionality, or the same ∅).

Position 3: operationality is derived, not ontological

Ontological status in VR-Sets belongs only to ∅. Operationality is a derived entity: it *is*, in the sense that it is given by its description, but it is not ontologically primary — it exists as an *action upon* ∅, not as a self-standing kind of being alongside ∅.

This distinction is principled. To say "we have two kinds of objects — ∅ and operationality" would be ontologically inaccurate: this would place operationality on the same level of primacy as ∅, violating minimalism. The precise formulation: ∅ alone is ontologically primary; operationality is present as a way of acting — action directed upon ∅ and upon the results of prior actions — and has no ontological status independent of this directedness.

A clarifying analogy. In arithmetic, the number 1 is ontologically primary; "add 1" is not an object but an action. One may speak of "the addition function" as a mathematical object — but this is already derivative from primary arithmetic, not a separate kind of number. So too in VR-Sets: ∅ is primary; operationality is action; "functionality as object" is derivative speech, not a separate ontological kind.

No atoms, individuals, labels, urelements, or ordered pairs as objects belong to the ontology of VR-Sets. The name of a set *is* the set itself (is operationality itself); not a separate syntactic pointer with ontological status. The slogan "only ∅ is, all else is doing" in application to this position means literally: only ∅ *is*; everything else *acts*, and action is not being.

Position 4: operational infinity in place of actual infinity

Infinity in VR-Sets is the **unbounded applicability of an operation**, not a "completed totality." The set ω is infinite not because it holds all natural numbers as a finished object, but because its functionality is not exhausted at any finite number of queries. The distinction is methodological: classical ZF postulates the existence of an actually infinite set; VR-Sets discovers operational infinity as a consequence of the closure principle (see Part III).

Position 5: operational set in place of set-as-container

A set in VR-Sets is an operational entity given by its membership functionality. A set *is* its functionality, not separate from it. This is the core definition, given formally in Part II. Here we fix it as an ontological position: "set = operationality" is not a renaming, but a replacement of the underlying metaphor.

The slogan

The slogan of the position, inherited from VR-Numbers Part VI.5 and applied to sets:

"Only ∅ is — all else is doing."

Applied to VR-Sets: a set *is* its operationality, not separate from it; ∅ is the sole ontologically primary object; all other sets are actions, not entities.

I.4. Reference vs. containment: the key distinction

The central distinction on which the resolution of self-reference paradoxes in VR-Sets rests is the distinction between membership as reference and membership as containment. We examine it through simple examples.

The container picture (classical ZF)

In classical ZF the set {a, b, c} is a "container" physically holding the elements a, b, c. The self-referential set A = {A} on this picture requires that A *contain itself as an element* — which generates the intuitive paradox: "how can a container be inside itself?" Russell's paradox is a sharpening of this difficulty.

Classical ZF resolves the paradox syntactically (axiom of separation: one cannot form a set from elements with an arbitrary property, only from elements of an already existing set with a property); type theory resolves it by stratification (a set and its element must lie at different levels). Both solutions preserve the container metaphor while imposing restrictions on it.

The reference picture (VR-Sets)

In VR-Sets a set is not a container. A set is a procedure that answers queries. $x \in y$ means: "the functionality y, in one of its responses, reveals x." This is *reference*: y uses x in its description.

The self-referential set $A = \{A\}$ on this picture: A is a functionality that, upon a query, reveals A. There is no "physical embedding of itself within itself": there is a procedure that returns a definite response. That this response happens to be A is no more paradoxical than a recursive function calling itself within its own definition. The structure is familiar to a programmer:

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function A() { return A; }
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— there is no contradiction here. A is a function whose only response is itself. This is an operationally defined description.

Where the difference lies

The distinction between the two pictures is not mathematical (the mathematics of the same sets remains the same) but **semantic**. The container picture reads $x \in y$ as a claim about spatial structure; the reference picture reads it as a claim about computational behaviour. If this were merely a renaming, the distinction would be empty. But it is substantive:

(a) **Russell's paradox** in the container picture requires syntactic restrictions (axiom of separation). In the reference picture it is dissolved ontologically: the description R = "reveal x if and only if $x \notin x$" demands a contradictory response to the query "$R \in R$?" and therefore does not specify a describable functionality. R is not "forbidden" — it *does not exist* as a functionality.

(b) **Self-membership** in the container picture is a pathology, requiring a separate axiom (anti-foundation) to be admitted. In the reference picture it is a natural possibility, restricted only by the structural condition of well-foundedness (Part IV). Whether to admit it or forbid it is an ontological choice, not a mathematical necessity.

(c) **Extensionality** in the container picture is a separate axiom (sets with the same elements are equal). In the reference picture it is a tautology of the identity definition: sets with the same responses to queries are the same functionality (see Definition 4 of Part II).

An analogy from programming

The distinction between reference and containment is routine in programming. Given a structure

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node = { value: 1, next: node }
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— this is a cyclic linked list. There is no "infinite embedding" in memory; there is a pointer node.next referring to node. The structure is well-defined, not pathological, not paradoxical. The programmer distinguishes "node holds next as a field" (a reference) from "node physically contains a copy of itself" (which would be an infinite matryoshka). VR-Sets makes the same distinction for sets: $A \in A$ is a pointer from the functionality A to itself, not an infinite matryoshka.

This analogy is substantive, not decorative. Modern approaches to non-well-founded sets (Aczel 1988, Barwise–Moss 1996) explicitly exploit graph representations in which cycles are admissible data structures. VR-Sets does the same on the ontological level.

I.5. Countability of the operational universe

A fundamental consequence of the operational ontology: the entire universe of sets in VR-Sets is countable. This is a sharp divergence from classical ZF, where the universe V is a proper class of uncountable cardinality.

The argument

Every set in VR-Sets is given by a *description* of its functionality. A description is a finite syntactic structure (even when the functionality reveals countably many elements — as with ω — the description itself is finite: "the n-th query yields O_n"). The set of finite descriptions over a finite alphabet is countable: this is a basic fact of the theory of formal languages (a countable union of finite sets is countable).

Therefore the operationally describable functionalities are countable in number. By the closure principle (Part II §5), every describable functionality is a set. Hence the sets in VR-Sets are countable in number.

Comparison with classical ZF

In classical ZF the sets form a proper class of uncountable cardinality, and $\wp(\omega)$ alone is already uncountable (Cantor's theorem). VR-Sets departs radically from this picture: $\wp(\omega)$ is countable in VR-Sets (Part III, §III.5), and this is not a defect but a consequence of the ontological choice.

Detailed discussion of this divergence is given in Part VI. Here we fix it as a fundamental fact: **the operational universe of VR-Sets is countable, and from this follow many structural features of the theory** — the absence of uncountability, the simplification of the axiom of choice, the absence of the Banach–Tarski paradox.

I.6. Terminological glossary

We fix here the terminology used consistently throughout all parts of the preprint.

Ontological terms

Operational set — a set in VR-Sets, given by its membership functionality.

Operationality — the membership functionality, that which is the set. Used as a synonym for "functionality" in contexts where the ontological status is at issue.

Functionality membership (or simply functionality) — a procedure responding to queries "what are your elements?"; either reveals a definite element or reveals nothing.

Operational infinity — unbounded applicability of an operation, contrasted with the actual infinity of classical ZF.

Operational depth — the least number of unfolding steps after which a functionality reaches ∅. The depth of ∅ is 0; finite sets have finite depth; ω has non-finite but non-cyclic depth; cyclic sets (such as the Quine atom) have no well-defined depth.

Operational identity ≡ — coincidence of functionalities: A ≡ B if every element of A matches an element of B under ≡, and conversely. The analogue of Leibnizian equality from VR.

A note on the sign ≡

The sign ≡ in VR-Sets denotes operational identity. It is **not** used in any of the standard alternative senses that this sign carries in mathematical literature:

- not congruence modulo n (as in number theory);
- not logical equivalence (as between formulas);
- not definitional equality (as in type theory).

Rather, ≡ is the coinductive identity between operational functionalities, defined in Part II Definition 4. It is structurally equivalent to bisimulation in the AFA literature (Aczel 1988), and readers familiar with that tradition may regard ≡ as bisimulation lifted to the operational setting. We use ≡ rather than = to signal explicitly that identity of sets in VR-Sets is defined through coincidence of functionalities, not through any pre-existing notion of equality.

Syntactic terms

Description of a functionality — a finite syntactic record specifying an operational procedure. A set is describable if it has a description.

Name of a set — the set itself. The ontology of VR-Sets contains no separate layer of "names denoting objects": the name and the named coincide.

Mode-related terms

ZFC-mode — the operational universe restricted to well-founded functionalities. Formally defined in Part IV.

ZFA-mode — the operational universe without the well-foundedness restriction; admitting self-reference and cycles.

Well-founded — a functionality is well-founded if every sequence of successive unfoldings of its responses terminates at ∅ in a finite number of steps. Discussed in detail in Part IV §IV.2.

Meta-principles

The closure principle — every operationally describable functionality is a set. Formal statement in Part II §3. In ZFC-mode the principle is applied subject to well-foundedness; in ZFA-mode without restriction.

Operational definiteness — a functionality returns a definite response to every query; contradictory descriptions (requiring contradictory responses) do not specify functionalities. This is what dissolves Russell's paradox.

I.7. Structure of the preprint

The preprint consists of nine parts, listed below. The present part is the first.

Part I. Foundations. Place in the cycle of works; the central idea; ontological position; the distinction between reference and containment; countability of the operational universe; terminological glossary; structure of the preprint.

Part II. The Operational Construction of Sets. Four core definitions (set, ∅, kinds of sets by operationality, operational identity ≡), three lemmas (extensionality, uniqueness of ∅, operational depth), the closure principle, the resolution of Russell's paradox, principal examples.

Part III. The ZF Axioms as Closure Theorems. All eight ZF axioms together with AC are closure theorems of the operational universe (or follow from the definitions of Part II). Divergences from classical ZF: $\wp(\omega)$ is countable; the replacement schema becomes a single theorem; foundation is a mode-dependent property.

Part IV. The Two Modes: ZFC and ZFA. Formal definition of the ZFC-mode and ZFA-mode through well-foundedness as a structural property of a functionality. Theorems of completeness of each mode relative to the corresponding classical theory. Conjecture of equivalence — via a countable model.

Part V. The Numbers of VR in VR-Sets. The von Neumann ordinals O_n and ω as specific well-founded operational sets. Automatic membership in the ZFC-mode as a theorem. Arithmetic inherited via isomorphism with VR Part I.

Part VI. Cardinality, Cantor, Choice. Countability of the operational universe in full detail. Cantor's diagonal argument in the operational ontology. AC as a theorem on the countable universe. Absence of the Banach–Tarski paradox by construction.

Part VII. Truth and Self-Reference. Tarski's theorem on the undefinability of truth in light of the operational ontology. The duality of axiom A1 and the hierarchy of levels of truth through applications of t. A cautious formulation: a reformulation, not a resolution.

Part VIII. Ontological Position. A parallel to VR-Numbers Part VI. "Only ∅ is, all else is doing" applied to sets, with detailed exposition of all ontological commitments.

Part IX. Open Questions. Connections with topoi, homotopy type theory (HoTT), formalisation in Lean, machine verification of equivalence conjectures, extensions (Morse–Kelley, NBG).

Each part can be read with minimal context from preceding ones, but Parts II–IV form the core of the system, on which the remaining parts rest.